

418-hw6

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Problem 1

```
# a
cov.mat = matrix(c(8,2,3,1,2,5,-1,3,3,-1,6,-2,1,3,-2,7),4,4,byrow=T)
r11 = cov.mat[1:2,1:2] # top left
r12 = cov.mat[1:2,3:4] # top right
r21 = cov.mat[3:4,1:2] # bottom left
r22 = cov.mat[3:4,3:4] # bottom right
e1 = solve(r11)%*%r12%%solve(r22)%*%r21
e2 = solve(r22)%*%r21%%solve(r11)%*%r12
# canonical correlations are square roots of non-zero eigenvalues of these
Eig1 = eigen(e1)
Eig2 = eigen(e2)
cat('Canonical Correlations:',sqrt(Eig1$values), '\n')
```

```
## Canonical Correlations: 0.5519301 0.489861
```

```
# b
Eig1$vectors
```

```
##           [,1]      [,2]
## [1,]  0.6583540 0.5452882
## [2,] -0.7527084 0.8382486
```

```
Eig2$vectors
```

```
##           [,1]      [,2]
## [1,]  0.9676678 0.5058921
## [2,] -0.2522283 0.8625968
```

```
# (U1,V1) = (0.6583*x1 - 0.7527*x2, 0.9676*x3 - 0.2522*x4)
# (U2,V2) = (0.5452*x1 + 0.8382*x2, 0.5058*x3 + 0.8626*x4)
```

Problem 2a

```
returns = read.csv('C:/Users/jacma/Downloads/Dataset_WeeklyRatesOfReturn.csv')
cor(returns)
```

```
##                JPMorganBank  CitiBank WellsFargoBank RoyalDutchShell
## JPMorganBank      1.00000000 -0.1922736      0.31321982      -0.1194831
## CitiBank          -0.19227360  1.00000000     -0.06523680       0.3731722
## WellsFargoBank    0.31321982 -0.0652368      1.00000000     -0.4111161
## RoyalDutchShell   -0.11948307  0.3731722     -0.41111605       1.0000000
## ExxonMobil        0.02614869  0.6852879     -0.01034666       0.1797010
##                ExxonMobil
## JPMorganBank      0.02614869
## CitiBank          0.68528795
## WellsFargoBank    -0.01034666
## RoyalDutchShell   0.17970100
## ExxonMobil        1.00000000
```

Negative correlation coefficients indicate that one company's return rate decreases as the other company's return rate increases, whereas positive coefficients indicate that the return rate increases for one if the other's rate increases. Thus, a negative value (as seen between Wells Fargo and Royal Dutch) could suggest that one company benefit from another experiencing decreased rates of return. That is, the companies may be rivals or competitors. On the other hand, a positive value can indicate a partnership or reliance of one company on the other, since they benefit from another company succeeding. This relationship is seen between CitiBank and ExxonMobil.

Problem 2b

```
covar = as.matrix(cov(returns)) # 1-3 are banks, 4 and 5 are oil companies
covar
```

```
##                JPMorganBank  CitiBank WellsFargoBank RoyalDutchShell
## JPMorganBank      3.39689902 -1.102139      4.30555484      -2.0782852
## CitiBank          -1.10213937  9.672775     -1.51323628       10.9532322
## WellsFargoBank    4.30555484 -1.513236     55.62591164     -28.9374642
## RoyalDutchShell   -2.07828525  10.953232    -28.93746421      89.0666120
## ExxonMobil        0.02720391  1.203064     -0.04355907       0.9572997
##                ExxonMobil
## JPMorganBank      0.02720391
## CitiBank          1.20306366
## WellsFargoBank    -0.04355907
## RoyalDutchShell   0.95729973
## ExxonMobil        0.31862508
```

```
R.11 = covar[1:3,1:3]
R.12 = covar[1:3,4:5]
R.21 = covar[4:5,1:3]
R.22 = covar[4:5,4:5]
E.1 = solve(R.11)%*%R.12%*%solve(R.22)%*%R.21
```

```
E.2 = solve(R.22)%*%R.21)%*%solve(R.11)%*%R.12
Eigen1 = eigen(E.1) # additional eigenvalue, 3x3 matrix -> 3 eigenvalues
Eigen2 = eigen(E.2) # 2x2 matrix -> 2 eigenvalues, first two equivalent to above
cat('Canonical Correlations:',sqrt(Eigen1$values), '\n')
```

```
## Canonical Correlations: 0.7591317 0.3827057 6.184273e-09
```

```
# The canonical correlations are correlations between linear combinations of the
# original variables. Since the maximum dimension of the subsetted covariance
# matrix is 3, there are three possible canonical variables, and thus three
# possible canonical correlations. The highest resulting correlation was 0.759,
# which corresponds to the linear combination which most highly correlates the
# return rates of the banks with the return rates of the oil companies. This is
# the best possible option as it best explains the relationship between the sets
# of variables. The second canonical correlation does not explain as much of the
# relationship, while the third explains almost none. The very small third value
# was to be expected because the first combination would have no countering
# combination with which to explain the overall correlation.
```

Problem 2c

```
Eigen1$vectors
```

```
##           [,1]      [,2]      [,3]
## [1,] 0.3632271 0.2562083 0.99637232
## [2,] 0.9262482 0.5164139 -0.02547016
## [3,] -0.1006498 0.8171132 -0.08120015
```

```
Eigen2$vectors
```

```
##           [,1]      [,2]
## [1,] -0.03198232 -0.09134972
## [2,] -0.99948843 0.99581887
```

```
# (U1,V1) = (0.3632*JPMorgan + 0.9262*Citibank - 0.1006*WellsFargo,
#           -0.0319*RoyalDutch - 0.9994*ExxonMobil)
# (U2,V2) = (0.2562*JPMorgan + 0.5164*Citibank + 0.8171*WellsFargo,
#           -0.0913*RoyalDutch + 0.9958*ExxonMobil)
```